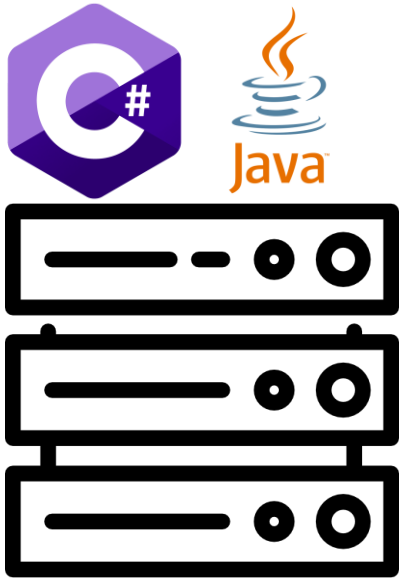
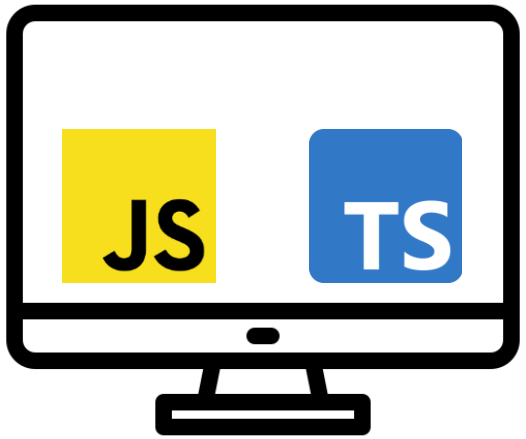
Blazor

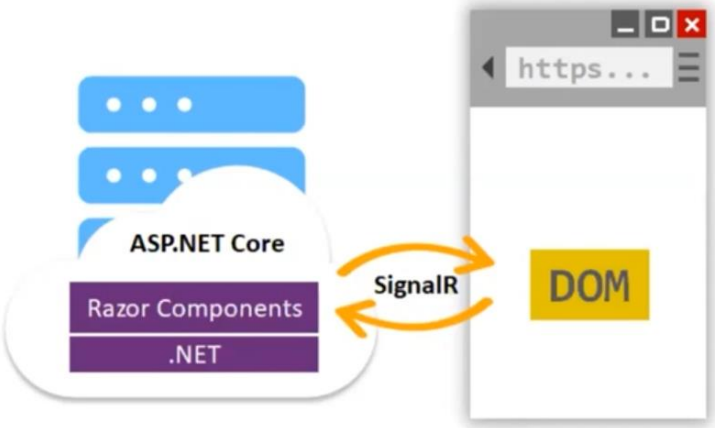


фулстек веб платформа для SPA

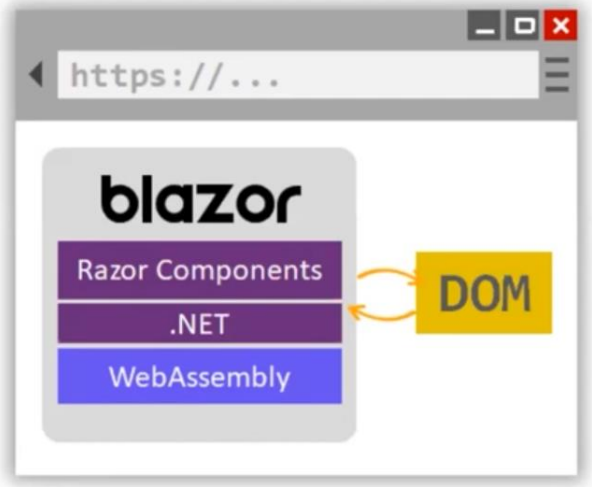


можно назвать аналогом

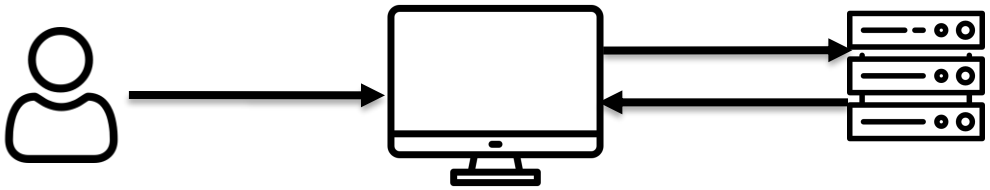
Blazor Server



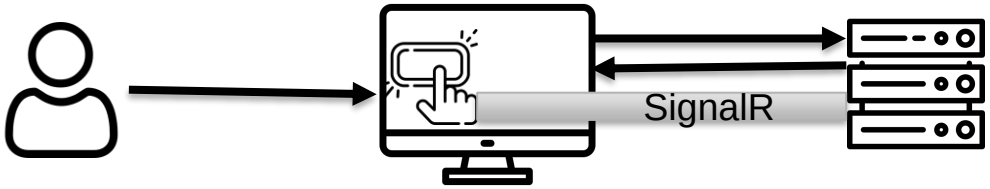
Blazor Web Assembly



None (Server Side Render Mode)

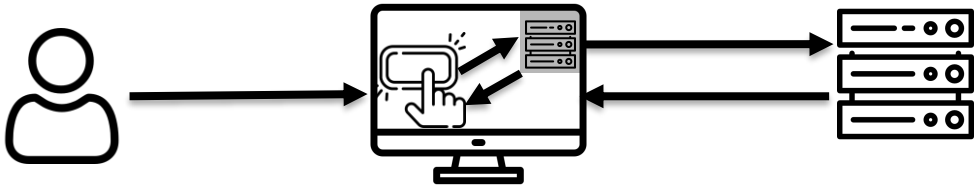


Blazor Server

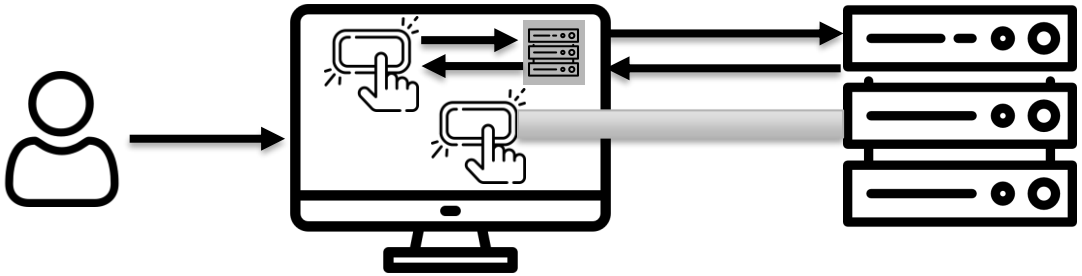


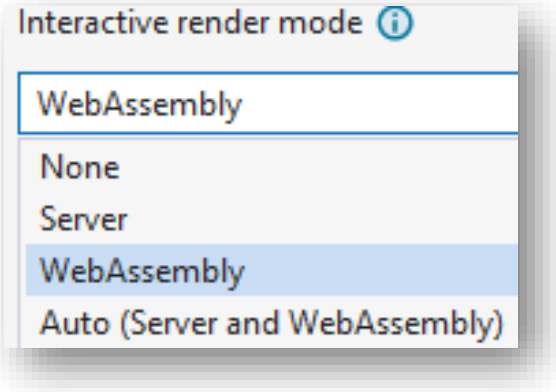
Blazor Web Assembly

WASM

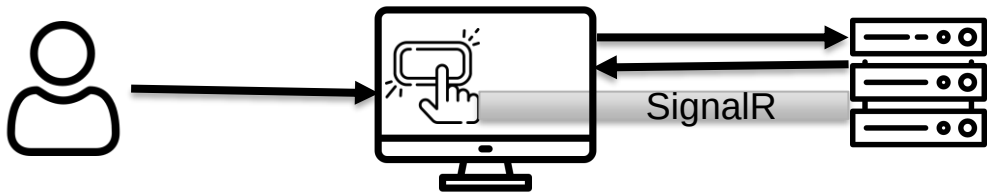


Combined Mode

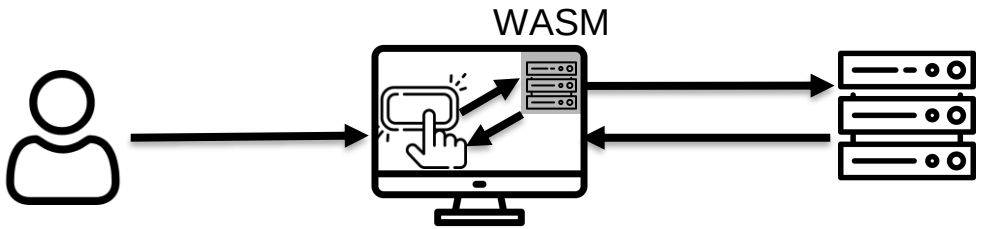




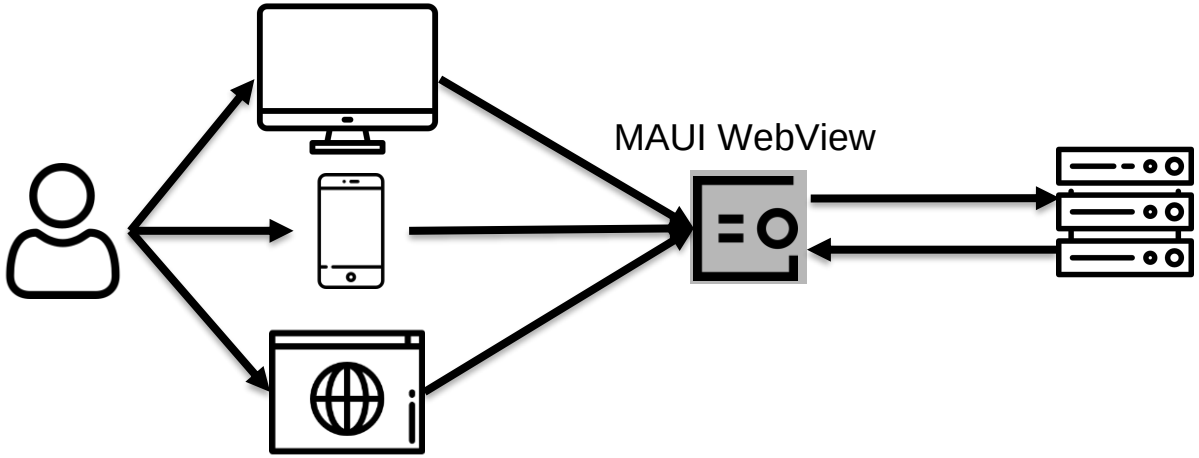
Blazor Server



Blazor Web Assembly



Blazor Hybrid



Blazor Web Assembly

- ✓ Compiles to static files
- ✓ Offline state support
- ✓ Codesharing between client and server
- − First time payload
- − Load time on poor connections
- − Restricted runtime
- − Security restrictions

Blazor Server

- ✓ Full .Net Core runtime
- ✓ Fast development cycle
- ✓ Small download size
- ✓ Code is kept on server-side
- − Issues in high latency environments
- − No offline support
- − Heavy resource demand on the server

Interactivity location

Global

Per page/component

Global

Counter.razor | App.razor

BlazorApp2.Client | BlazorApp2.Client.Pages.Counter | currentCount

```razor
@page "/counter"
@rendermode InteractiveWebAssembly

<PageTitle>Counter</PageTitle>

<h1>Counter</h1>

<p role="status">Current count: @currentCount</p>

<button class="btn btn-primary" @onclick="IncrementCount">Click me</button>

@code {
    private int currentCount = 0;

    private void IncrementCount()
    {
        currentCount++;
    }
}
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'BlazorApp2' (2 of 2 projects)
  - BlazorApp2
    - Connected Services
    - Dependencies
    - Properties
    - wwwroot
    - Components
      - Layout
      - Pages
        - _Imports.razor
        - App.razor
        - Routes.razor
    - appsettings.json
    - Program.cs
  - BlazorApp2.Client
    - Connected Services
    - Dependencies
    - wwwroot
    - Pages
      - Counter.razor
      - _Imports.razor
    - Program.cs

Counter.razor | App.razor

BlazorApp2

```razor
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <base href="/" />
    <link rel="stylesheet" href="bootstrap/bootstrap.min.css" />
    <link rel="stylesheet" href="app.css" />
    <link rel="stylesheet" href="BlazorApp2.styles.css" />
    <link rel="icon" type="image/png" href="favicon.png" />
    <HeadOutlet />
</head>

<body>
    <Routes />
    <script src="_framework/blazor.web.js"></script>
</body>

</html>
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'BlazorApp2' (2 of 2 projects)
  - BlazorApp2
    - Connected Services
    - Dependencies
    - Properties
    - wwwroot
    - Components
      - Layout
      - Pages
        - _Imports.razor
        - App.razor
        - Routes.razor
    - appsettings.json
    - Program.cs
  - BlazorApp2.Client
    - Connected Services
    - Dependencies
    - wwwroot
    - Pages
      - Counter.razor
      - _Imports.razor
    - Program.cs

Counter.razor | App.razor | Program.cs

BlazorApp1.Client | BlazorApp1.Client.Pages.Counte | currentCount

```razor
@page "/counter"

<PageTitle>Counter</PageTitle>

<h1>Counter</h1>

<p role="status">Current count: @currentCount</p>

<button class="btn btn-primary" @onclick="IncrementCount">Click me</butto

@code {
    private int currentCount = 0;

    private void IncrementCount()
    {
        currentCount++;
    }
}
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'BlazorApp1' (2 of 2 projects)
  - BlazorApp1
    - Connected Services
    - Dependencies
    - Properties
    - wwwroot
      - bootstrap
      - app.css
      - favicon.png
    - Components
      - Pages
        - Error.razor
        - _Imports.razor
        - App.razor
    - appsettings.json
    - Program.cs
  - BlazorApp1.Client
    - Connected Services
    - Dependencies
    - wwwroot
      - appsettings.json
      - appsettings.Development.json
    - Layout
    - Pages
      - Counter.razor

Counter.razor | App.razor | Program.cs

BlazorApp1

```razor
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0"
    <base href="/" />
    <link rel="stylesheet" href="bootstrap/bootstrap.min.css" />
    <link rel="stylesheet" href="app.css" />
    <link rel="stylesheet" href="BlazorApp1.styles.css" />
    <link rel="icon" type="image/png" href="favicon.png" />
    <HeadOutlet @rendermode="InteractiveWebAssembly" />
</head>

<body>
    <Routes @rendermode="InteractiveWebAssembly" />
    <script src="_framework/blazor.web.js"></script>
</body>

</html>
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

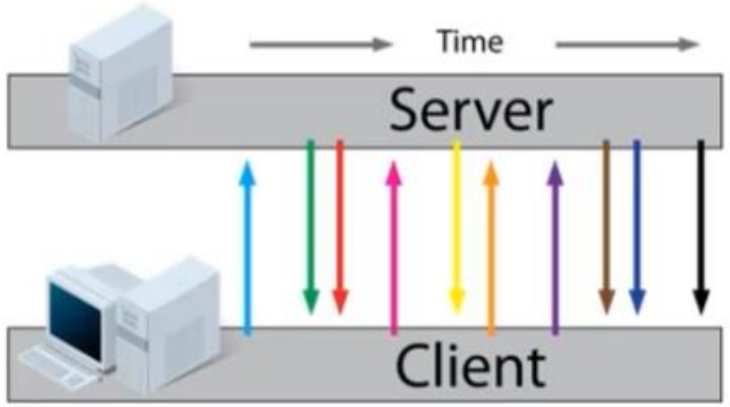
- Solution 'BlazorApp1' (2 of 2 projects)
  - BlazorApp1
    - Connected Services
    - Dependencies
    - Properties
    - wwwroot
      - bootstrap
      - app.css
      - favicon.png
    - Components
      - Pages
        - Error.razor
        - _Imports.razor
        - App.razor
    - appsettings.json
    - Program.cs
  - BlazorApp1.Client
    - Connected Services
    - Dependencies
    - wwwroot
      - appsettings.json
      - appsettings.Development.json
    - Layout
    - Pages
      - Counter.razor

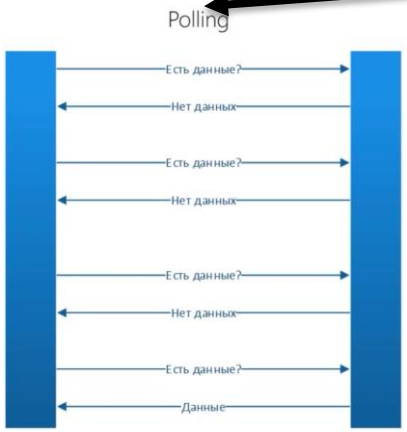
Синхронная модель

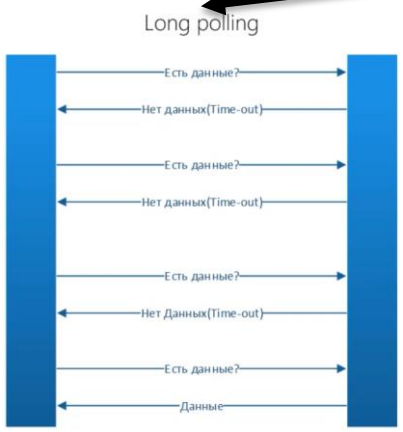
SignalR – это

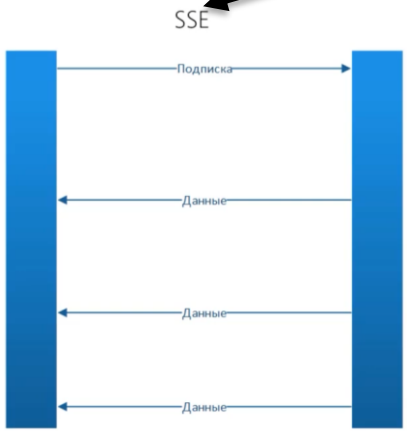
- библиотека, которая упрощает процесс добавления веб-функций в режиме реального времени в приложения
- абстракция транспорта, необходимого для работы в режиме реального времени между клиентом и сервером

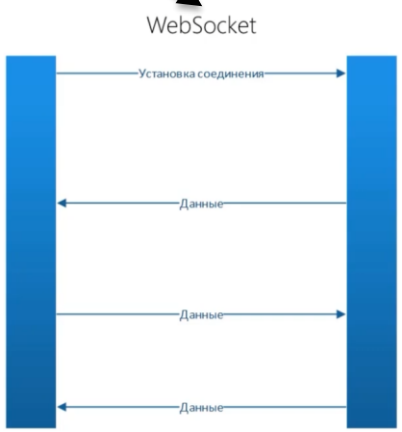
Модель реального времени (асинхронная)











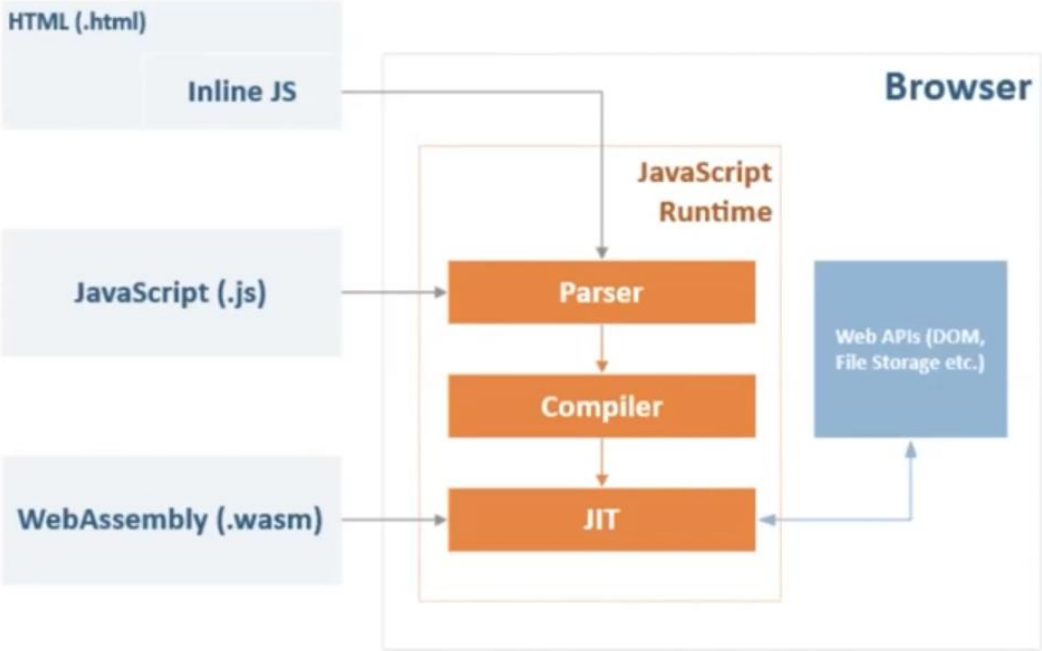
https://github.com/appcypher/awesome-wasm-langs

- Stable for production usage.

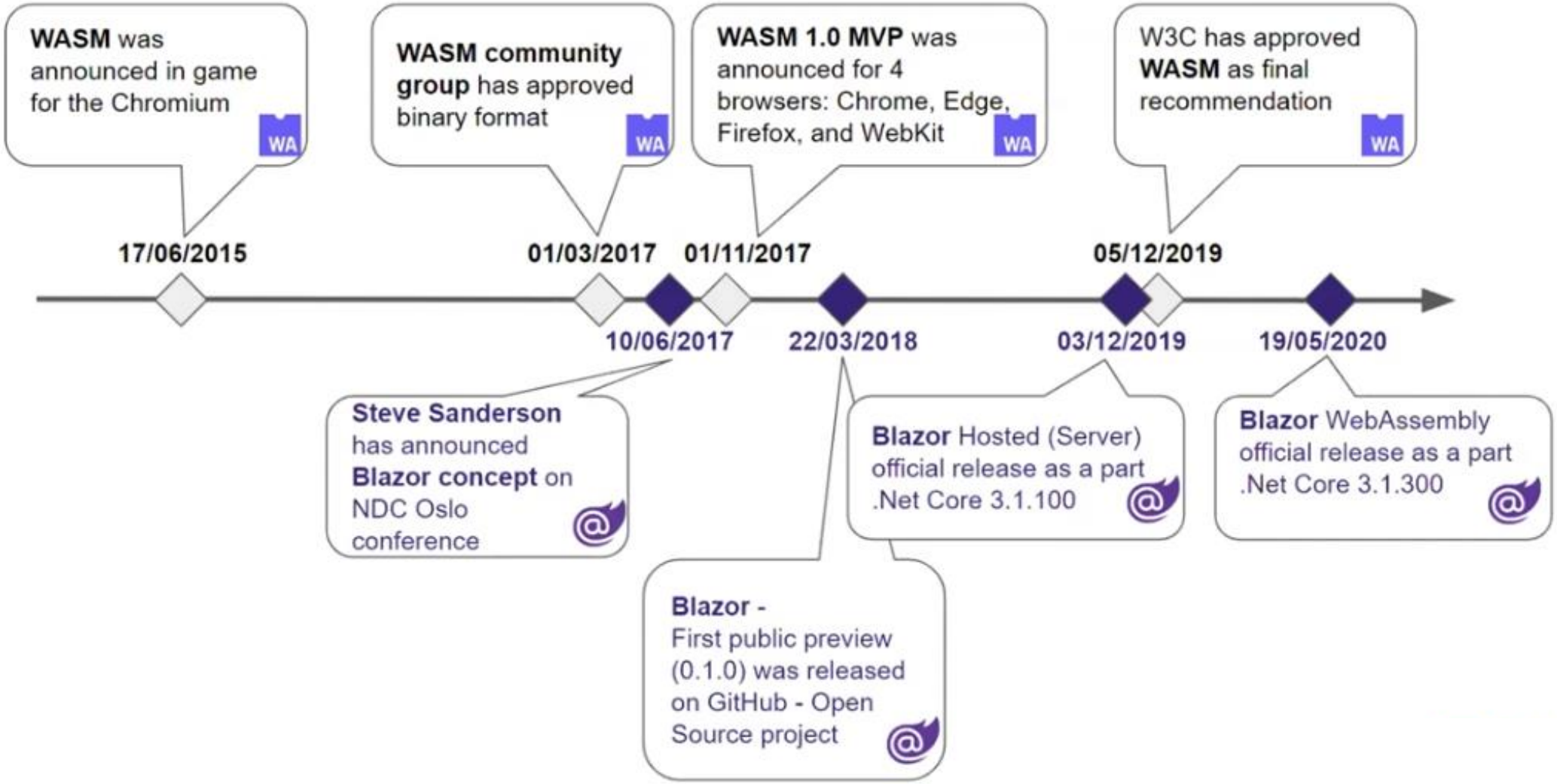
- .Net
- AssemblyScript
- Brainfuck
- C#
- Clean
- Cyber
- COBOL
- F#
- Forth
- Grain
- LabVIEW
- Lobster
- Lox
- Lua
- Nelua
- Never
- Rego
- TypeScript
- WebAssembly
- Zig

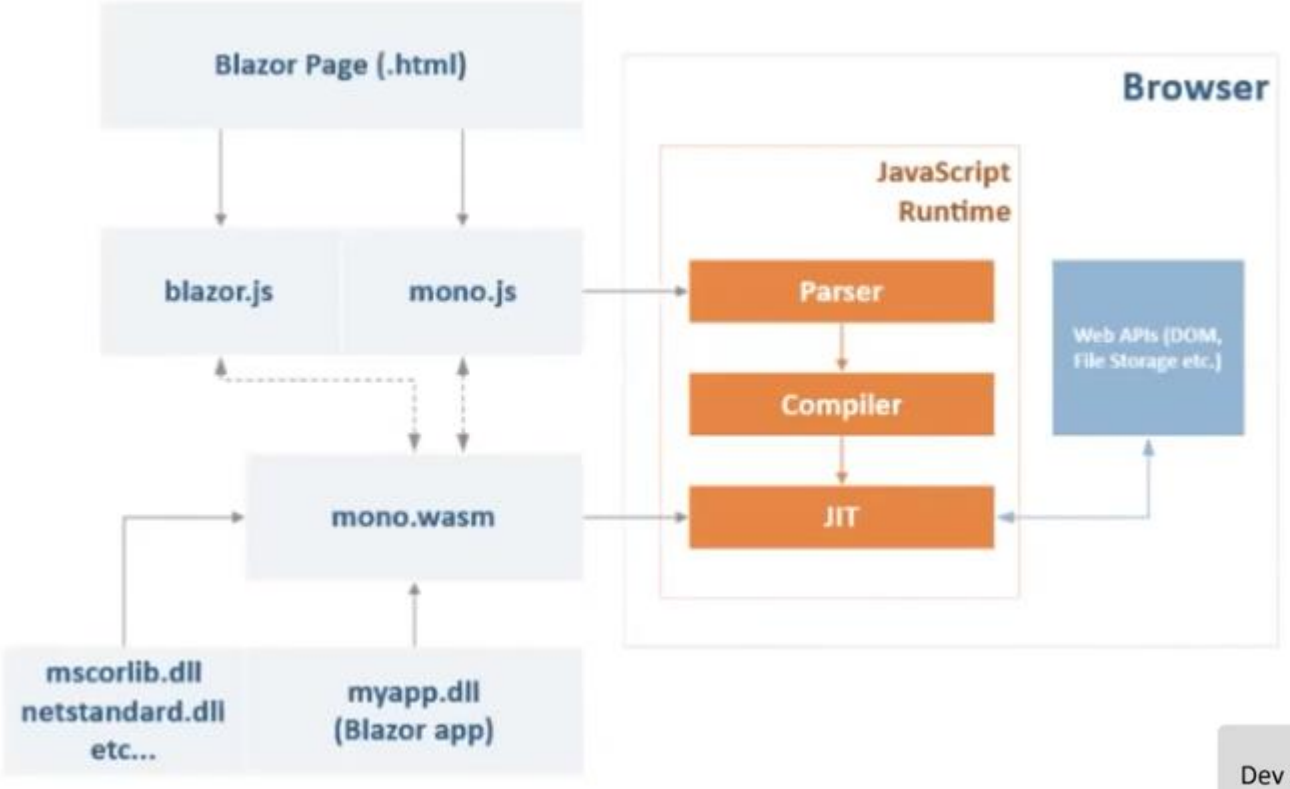
WebAssembly – это

- Бинарный формат, позволяющий запускать код в браузере
- Низкоуровневая виртуальная машина
- Результат компиляции с языка более высокого уровня



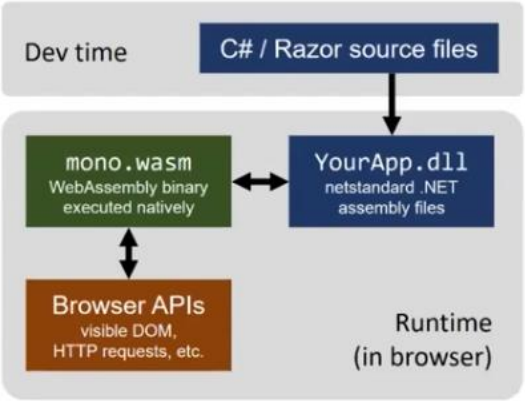
WebAssembly & Blazor timeline





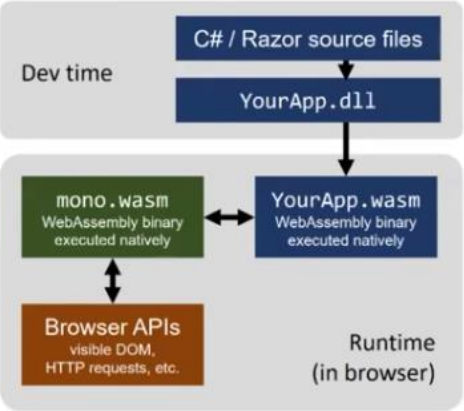
до .NET 6

Interpretation



начиная с .NET 6

AOT



Razor – это синтаксис разметки для внедрения кода на основе .NET на веб-страницы.

```
@page "/counter"
@rendermode InteractiveWebAssembly

<PageTitle>Counter</PageTitle>

<h1>Counter</h1>

<p role="status">Current count: @currentCount</p>

<button class="btn btn-primary" @onclick="IncrementCount">Click me</button>

@code {
    private int currentCount = 0;

    private void IncrementCount()
    {
        currentCount++;
    }
}
```

Синтаксис razor pages состоит из:

- Razor markup
- HTML
- C#

Расширение файлов:

- *.razor – компоненты и страницы для Blazor
- *.cshtml – только страницы